

Leonardo De Andrade Santos

Software / Embedded Systems Engineer | Realtime Systems • C/C++ • Python • FPGA/VHDL • DSP

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Professional Summary

Software and embedded systems engineer with 4+ years of experience building production C/C++ and Python systems close to hardware, plus FPGA/VHDL research in deterministic DSP datapaths. Strong fit for realtime quantum-control software: comfortable reasoning about latency, jitter, fixed-point arithmetic, hardware/software boundaries, simulation-based validation and measured behaviour on target systems. Master's research focuses on RF power-amplifier modelling, Digital Predistortion and FPGA-oriented implementation of mathematically defined algorithms.

Technical Skills

Systems Software: C/C++, Python, Linux, shell scripting, embedded Linux, realtime embedded systems, hardware/software integration, performance debugging

Realtime & Hardware Interfaces: deterministic processing, latency-aware design, sensor/data acquisition pipelines, communication interfaces, telemetry systems, device integration, bare-metal concepts

FPGA, RTL & DSP: VHDL, RTL simulation, fixed-point arithmetic, pipelined datapaths, complex arithmetic, delay lines, DSP for RF systems, model-to-RTL validation

Algorithms & Verification: Python/NumPy modelling, reference models, bit-accurate comparison, testbenches, simulation replay/validation, Git, code review, CI/CD-oriented workflows

Tools: Linux, Git, debugging, Xilinx ISE, Vivado academic exposure, GHDL/ModelSim exposure, Cadence Genus/ASIC flow exposure

Professional Experience

Embedded & Software Engineer — IoTaq Technology

Oct 2022 – Feb 2026

- Developed low-level C/C++ software for embedded and Linux-based systems used in telemetry, equipment monitoring and field diagnostics.
- Built data acquisition and communication pipelines between devices, sensors and backend services, with attention to reliability under deployed hardware constraints.
- Debugged hardware/software integration issues, communication failures and performance problems across embedded devices and production systems.
- Created Python tooling for analysis, validation and reliability checks over equipment datasets and telemetry outputs.
- Worked across firmware, Linux systems and backend integration, translating hardware behaviour into software requirements and operational tooling.

Engineering Intern — Lactec

2022

- Developed and tested embedded measurement systems and capacitive sensor prototypes, including validation, data acquisition and hardware/software integration.

Realtime, FPGA & Research Projects

Realtime DSP and FPGA Implementation for RF Power Amplifier Linearization

- Designed and evaluated Digital Predistortion algorithms for RF amplifier linearization using memory polynomial models.
- Implemented Python reference models and translated selected structures into fixed-point VHDL datapaths for FPGA-oriented execution.
- Built validation flows comparing Python fixed-point outputs with VHDL simulation results, focusing on bit-accurate behaviour, numerical precision and NMSE performance.
- Worked with complex arithmetic, coefficient quantization, delay-line memory, polynomial terms, pipelining and resource/latency trade-offs.
- Measured implementation characteristics including clock frequency, latency, resource use and simulation timing, rather than relying only on algorithm-level estimates.

FPGA-based TRNG / ASIC Research

- Worked on a true random number generator described in VHDL, including RTL understanding, modification and validation in a microelectronics research context.

DSP and Wireless Systems Research — GICS / UFPR

- Conducted research in wireless signal processing, RF optimisation and hardware-aware DSP implementation; presented research at Seminários de Microeletrônica do Paraná.

Baja SAE — Electrical and Embedded Systems

- Designed wiring harnesses and PCB-related subsystems; developed firmware and supported integration testing between electrical, mechanical and embedded subsystems.

Education

Master of Electronic Engineering — Universidade Federal do Paraná (UFPR), Brazil

Expected Aug 2026

Research focus: RF power amplifier behavioural modelling, Digital Predistortion, memory polynomial models, realtime DSP systems and FPGA/VHDL implementation.

Bachelor of Electronic Engineering — Universidade Federal do Paraná (UFPR), Brazil

Relevant background: digital systems, microelectronics, embedded systems, signal processing, firmware, PCB design and electrical systems.